Measuring Trace Resistance and Current Limit Testing

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10-4-2024

Introduction

Lab 11 and 12 consists of one board, and from this point on will be referred to as a single lab.

The purpose of this lab is to learn the best practices for measuring trace resistance, how to estimate resistance, test the maximum amount of current that different trace widths can handle and how voltage across each trace behaves. To do this, the resistances of different 1 inch long traces with different widths will be calculated by hand and measured using the typical 2-wire method and 4-wire method. The measurements will then be compared to the calculations to determine the best practice.

Next, the maximum current for the traces will be estimated using the Saturn PCB tool. This value will be compared against the true breakdown current (i.e. the current at which the trace burns) which will be determined experimentally. An increasing amount of current will be driven through a trace while monitoring the voltage read across it to examine how the properties of the trace changes. The current will be increased until the trace breaks down.

Plan of Record

What does it mean to work?

Since we are only studying the characteristics of traces using the board we are given, no board will be created for this lab. Therefore, the functionality of the board will be judged with different criteria:

1. Measured resistance should be close to the estimated resistance using the 4-wire method
2. The voltage reading across each trace should ramp up as the current running through increases past the recommended current.
3. A clear relationship between trace width and resistance should be derived from measurements
4. A clear relationship between voltage drop and current across the trace should be derived from measurements.

Features of the board:

1. 4 sets of 6-mil, 10-mil, 20-mil and 100-mil wides separated by a wide trace in the middle
2. 2 probe points at the end of each trace and 1 probe point at each end of the middle trace

Risk Reduction:

1. Safety goggles

Key Measurements:

1. Resistance <<1 ohm across each trace when supplying 1 amp via 4-wire method
2. Voltage drop across each trace with increasing current

Components:

1. Lab 11/12 trace board x1
2. Mini-grabber wires x4

Preliminary Calculations

Fig 1: Lab 11/12 board

To get a set of baseline values for each trace resistance, we first derive a general equation relating resistance, width and length.

Assuming 1oz copper thickness, $\rho = 1.68$ mohm-cm, $t = 34\mu m$

$$R = \rho \frac{\text{Length}}{A} = \frac{\rho(\text{length})}{(t \times w)} = \frac{\rho}{t \times w}(\text{length}) = R_{sq} \cdot n = .5 m\Omega \cdot n$$

Fig 2: Derivation of trace resistance equation

As long as the trace maintain constant properties (i.e. width, thickness and material), the equation relating its resistance to its dimensions can be reduced to simply

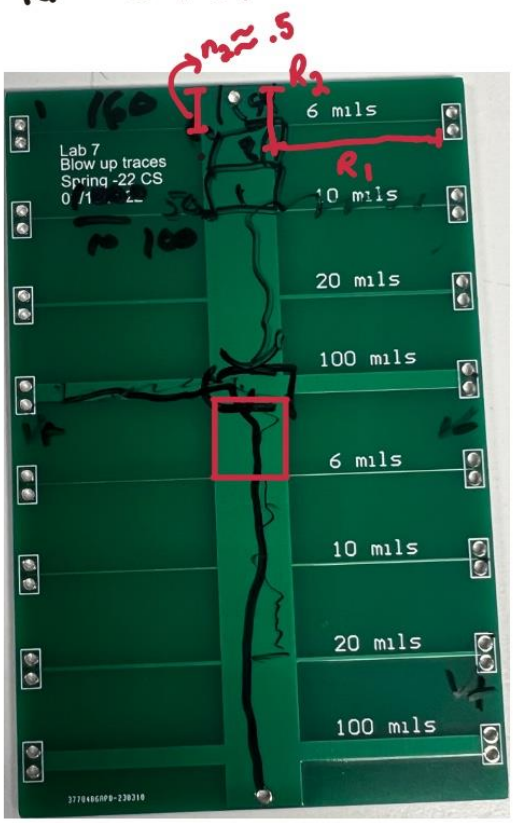
$$R = R_{sq} * n$$

$$R = .5\ m\Omega * n$$

Where R_sq refers to the resistance of a single square trace and n refers to the amount of square in a trace. Since the traces on this board are 1oz copper traces, R_sq is just .0005-ohm in all cases.

After deriving the general equation, the resistances of all of the traces were calculated.

for 6 mil:



$$R_1 + R_2 = (R_{sq} \cdot n_1) + (R_{sq} \cdot n_2)$$
$$= \left(.5 \cdot \frac{1\,in.}{6\,mil}\right) + (.5 \cdot .5)$$
$$= 80\,m\Omega + .25\,m\Omega$$
$$= \underline{80.25\,m\Omega}$$

Fig 3: Trace Calculation Procedure

NOTE: The black markings on the board do not pertain to the calculation of trace resistance

The total trace resistance can be broken down into two portions: The resistance due to the 1 inch portion, and the resistance due to the large middle trace. Because of this, each trace resistance were calculated separately and summed together. For the sake of simplicity, the distance between each trace was estimated to be one large trace square (red square in Fig. 3), for the 6 mils case, the distance from the trace to the contact point was estimated to be .5 of a square.

Following this procedure, the rest of the resistances were calculated

10 mil

$R = R_1 + R_2$

$= .5(100) + .5(1.5)$

$= 50 + .75 = 50.75\ m\Omega$

20 mil

$R = R_1 + R_2$

$R = .5(50) + .5(2.5) = 21.25\ m\Omega$

100:

$R = R_1 + R_2$

$= .5(10) + .5(3.5)$

$= 5 + 1.75 = 6.75$

Fig 4: Trace resistance calculations

Measuring Resistances: 2-Wire Method

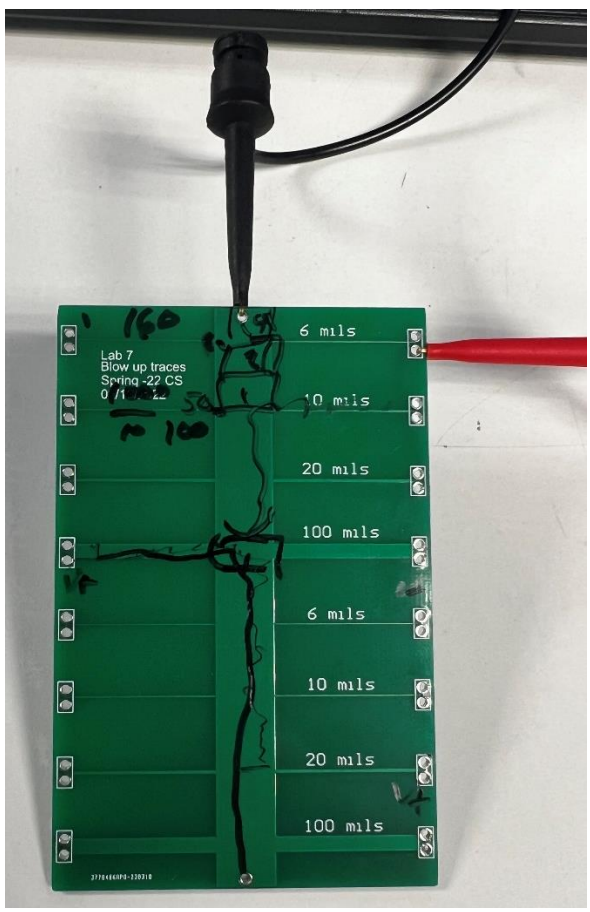


Fig 5: 2-wire method setup

To measure the resistances using the 2-wire method, a DMM was used with two leads, the ground lead connected to the top of the board and the positive lead connected to the trace of interest.

Measuring Resistances: 4-Wire Method

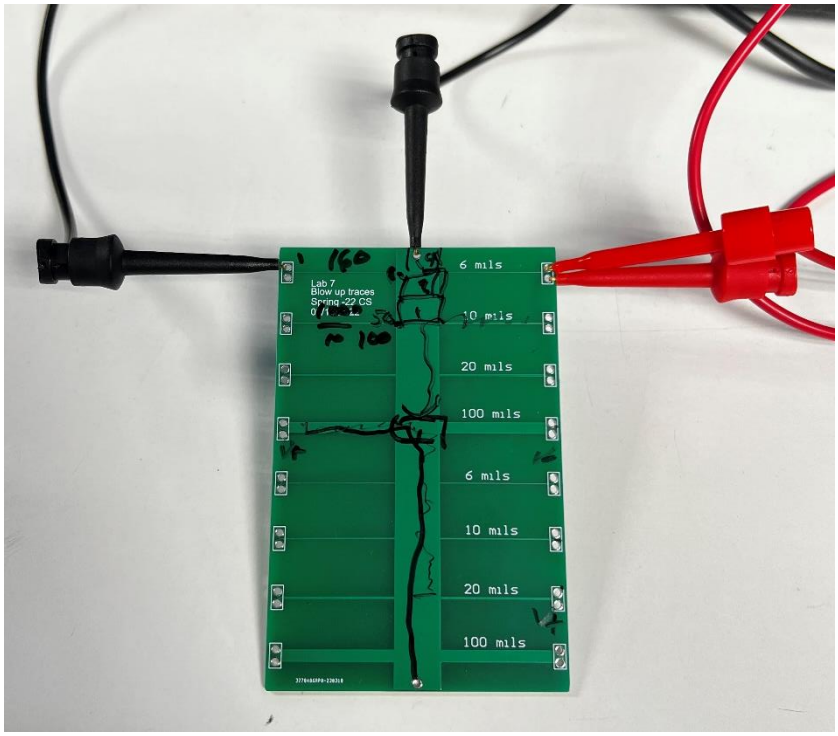


Fig 6: 4-wire method setup

To measure the resistance using the 4-wire method, a DC generator was used to send 1 amp of current through the trace of interest while a DMM was used to measure the voltage read on the wire.

The positive lead of both the DMM and DC generator was attached at the trace of interest while the ground lead of the generator was attached to the top of the board and the ground lead of the DMM was attached opposite of the positive leads. Since the traces are shorted, this setup allows for the voltage reading of just the trace of interest.

Measured Resistances

Line Width	Trace Square Number	Estimate (mΩ)	Center Square Number	2-Wire (mΩ)	4-Wire (mΩ)
6 mil	166	83.6	.5	210	76.39
10 mil	100	50.75	1.5	180	41.38
20 mil	50	26.25	2.5	150	19.63
100 mil	10	6.75	3.5	130	4.13

The 2-wire method consistently gave much higher readings than the expected values while the 4-wire method gave us values closer to what we expect, implying that for measuring resistances smaller than 100 mΩ, the 4-wire method should be used. These findings will be further elaborated on in the conclusion.

Current Limit Testing

The Saturn PCB toolkit was used to get an estimate of the maximum recommended current through a trace

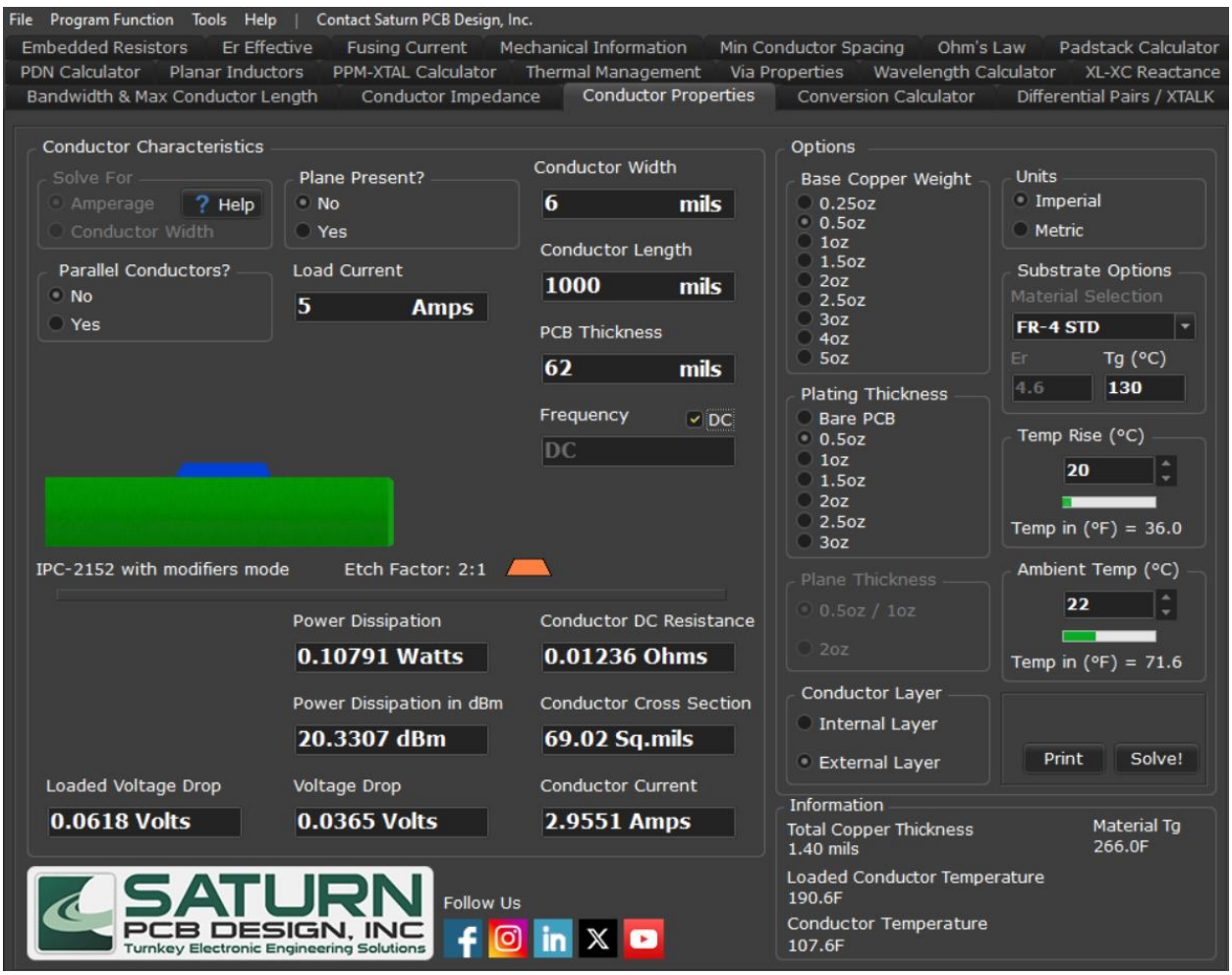


Fig 7: Saturn PCB Toolkit UI

Recommended Current Ratings

Line Width	Max Current Estimation (A)
6 mil	.811
10 mil	1.123
20 mil	1.715
100 mil	4.436

Once an estimate was gathered for each trace, we can compare them to actual measurements.

The configuration was the same as for the 4-wire resistance measurement method (figure 6). Current was incrementally stepped by .25 amps and was held at that current until the voltage reading stabilized. The current was stepped this way until either breakdown was achieved or 10 amps was reached (the current limit for the DC generator).

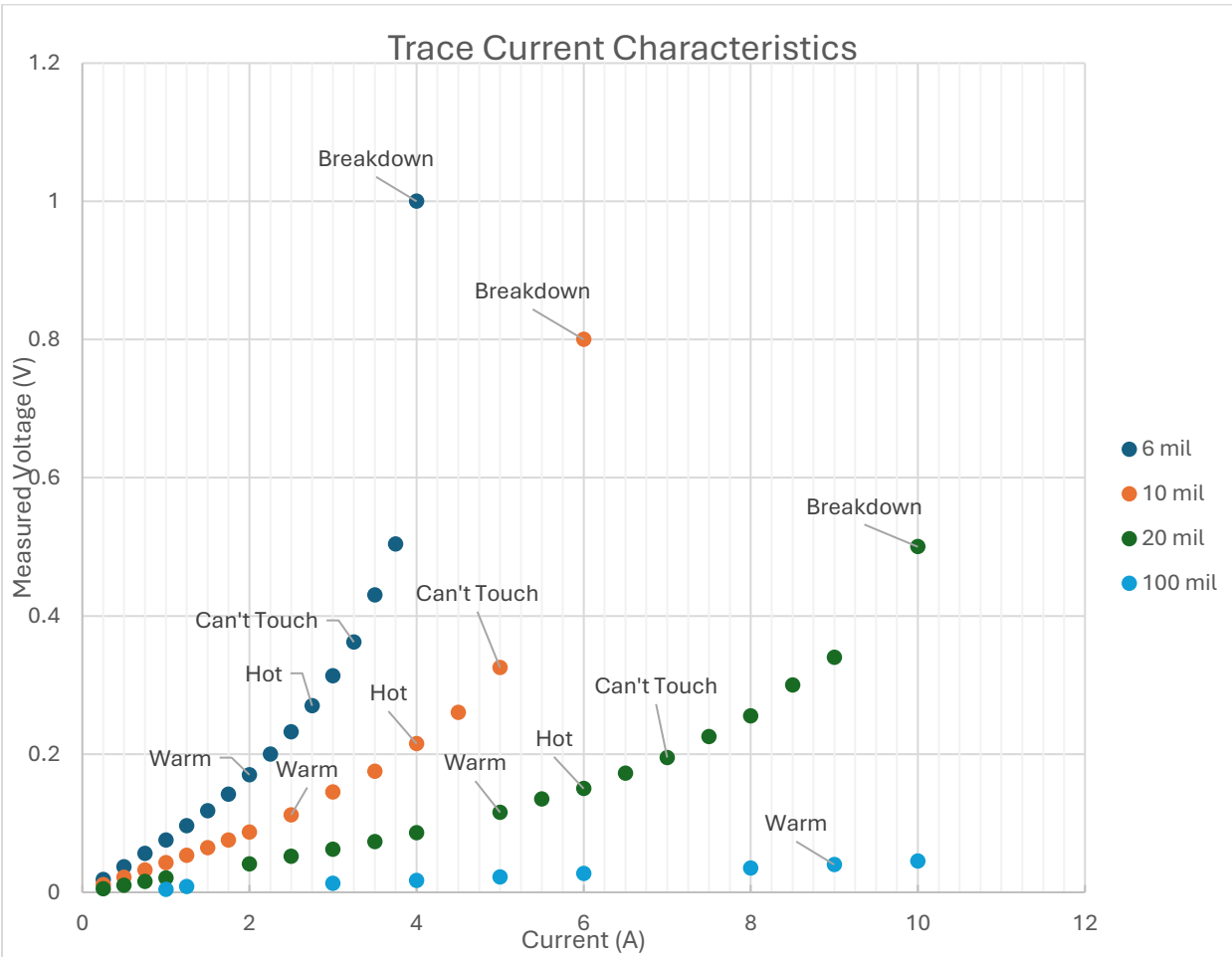


Fig 8: Trace current characteristics for 6, 10 20 and 100 mil trace widths

While current was being increased, the temperature was also being monitored. According to figure 8, the traces began warming up at around 1/3 of their apparent maximum temperature. Incidentally, the point at which the traces began to warm up was the same point that the voltage across the trace began to increase steadily over time rather than quickly stabilizing. This is also where the relationship between the current and voltage across the traces exhibits a non-linear relationship.

This is due to thermal runaway, where the increase in heat causes the resistance of the trace to increase.

Line Width	Estimated Current Limit (A)	Thermal Runaway Current (A)	True Max Current (A)
6 mil	.811	2	4
10 mil	1.123	2.5	6
20 mil	1.715	3	10
100 mil	4.436	9	N/A

We can see that the estimated current limits are well under the true maximum current and although a limit was estimated for the 100 mil trace, the maximum current was never reached. Further analysis will be covered in the conclusion.

Conclusion and Key Takeaways

To reiterate, the purpose of this lab was to learn the best practices for measuring small resistances as well as understand the current characteristics of different trace widths.

From this lab it is apparent that if there is any reason to read the resistances of traces, for example, if there were very long traces present in a PCB, the 4-wire method should be used. However, since it requires more resources and setup, it is sufficient to estimate the resistance mathematically. Additionally, the 2-wire method is sufficient for measuring the resistance of any component that acts as a load.

As for the maximum current that a trace can handle, the current at which thermal runaway occurs is an important property. The maximum current estimation given by the Saturn PCB toolkit is likely to be the max current at which the trace will not experience thermal runaway.

The following are key takeaways from this lab:

1. The 4-wire method should be used if there is need for exact trace measurements. In all other cases, the 2-wire method and estimations will be sufficient. This way, unnecessary time and energy won't be spent for resistance calculations

2. For circuit board designs, the current running through any trace should be estimated and kept note of to ensure that the current stays under the current recommendation given by the toolkit to avoid thermal runaway. This way, the voltage changes in the traces won't affect the rest of the board nor will it affect voltage readings for troubleshooting purposes.

3. Traces can handle much more current than the recommended amount. As long as the circuit is not designed in a way that subjects the traces to high current for long periods of time, they will do fine in cases of sudden current increases. In most cases, trace breakdown does not need to be factored into risk management.